

JONATHAN MA

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Education

Cornell University, College of Engineering (Ithaca, NY)

Master of Engineering, Systems Engineering

Anticipated December 2026

Bachelor of Science, Computer Science (GPA 3.99, USRA National Aerospace Scholarship)

Anticipated May 2026

- Coursework: Systems Programming, OS, Intro to Robotics, Computer Vision, Computer Graphics, AI/ML
- TA for Computer Systems (C, RISC V Assembly), OCaml Programming, and Intro to Robotics (ROS) classes

Experience

NVIDIA: Data Center Embedded Software Engineering Intern

May - August 2026

SpaceX: Flight Software Engineering Intern (Starshield)

May - August 2025

- Architected C++ flight and ground software solution spanning 5 satellite systems to deconflict satellite ops
- Conducted over 20 rigorous hardware-in-the-loop tests to ensure mission readiness with embedded Linux
- Organized design review and collaborated with 10+ stakeholders across teams, converting ambiguous guidelines to actionable tasks in order to develop a robust solution and testing plan
- Overhauled satellite integration and testing Python script to increase modularity and testability, taking ownership to drastically reduce risk for critical path test campaign while saving 1000s of engineering hours

Cornell Space Systems Design Studio: Alpha CubeSat Flight Software Lead

Oct 2022 - Present

- Promoted to lead of 5 person team after 2 yrs for exemplary performance and initiative to learn/solve problems
- Engineered real-time embedded C++ flight software for [1U CubeSat](#) & [light sail payload](#) on 6 mo LEO mission
- Developed code for memory constrained microcontroller with 32KB flash to interface with avionics firmware/drivers and control GPS, IMU, and LoRa radio peripherals via SPI, I2C, and UART
- Built and deployed [full-stack ground station](#) on Linux system with intuitive React and JavaScript control UI, Python Flask backend API, and Nginx reverse proxy to handle 100+ command & telemetry data points

RTX Collins Aerospace: Systems Engineering Intern (SEPP)

May - August 2024

- Collaborated with team of interns to conduct computer networking research studies using Info Centric Routing over IP with Raspberry Pi fleet, and extended 50K line open source C++ library to support network persistency

Johns Hopkins University Applied Physics Lab: Ground Software Engineering Intern

May - August 2023

- Led team of interns through design and creation of Angular and Java based app for parsing binary spacecraft command and telemetry packets for NASA's IMAP and Dragonfly missions, used by 50+ employees
- Applied Agile methodologies and solicited feedback from project leads throughout project duration

First Tech Challenge Robotics Team #7303: Robot Automation Lead

Aug 2019 - June 2022

- Won Maryland Tech Invite out of top 32 teams globally, Control Award for most innovative control/automation
- Collaborated w/ team to develop OpenCV perception, odometry localization, motion planning & PID control
- Developed multistage automation and CV algorithms which can autonomously generate paths to collect and shoot rings while moving at 5 in target from anywhere on a 12x12 ft field

Programming Projects

- Birdsong Synthesizer (C), Applied DSP direct digital synthesis algorithm to synthesize birdsongs on ARM-based Raspberry Pi 2040 microcontroller with DAC and ADC using the SPI protocol and GPIO pins
- [MirrorBot](#) (C, Python), Developed mirror control algorithm for robot using ROS, Arduino & Intel RealSense depth camera to detect faces and dynamically adjust reflections to encourage interactions in public spaces
- [Rubik's Cube Solving Robot](#) (C, Python), Arduino powered robot optimized to solve Rubik's Cube in 3-4 sec

Technical Skills

- Languages (Proficient): Python, C++, C, TypeScript, JavaScript, HTML/CSS, Java, OCaml, SQL, Verilog
- Frameworks: Arduino, ROS, OpenCV, Protothreads, NumPy, PyTorch, Pandas, Flask, React, Angular, Firebase
- Tools: Git, Docker, Linux, Bash, PlatformIO, TCP/IP, Elasticsearch, Kibana, Jira, CI/CD, BabylonJS/OpenGL